Name:

Exercise: Gesture → Interaction → Output

Choose one gesture or physical action for the basis of hypothetical interactive audiovisual work. Work backwards from what you want the interaction to produce, and consider what information the system needs to detect, measure, and map in order to realize the interaction.

You don't need to know exactly how to implement everything yet. Use the Technical Guide and Max Patches as references. Highlight anything you're unsure about.

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1. THE INTERACTION

What is the gesture or action?

Describe one physical action you or a participant might perform.

- *Example: Moving your hand from left to right in front of a camera.*

What should happen in response?

What would you like the system to do when it detects this action?

- *Example: A sound moves from left to right across the stereo field.*

What kind of feedback should the participant experience?

- ☐ Continuous (changes happen while the gesture is happening)
- ☐ Momentary (an action triggers once when the gesture is detected)
- ☐ Persistent (something remains active until another action occurs)
- ☐ Other: _____

Describe what the feedback might look like:

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2. WORK BACKWARDS: WHAT DO YOU NEED TO MEASURE?

Think about what information the system might need to extract from the gesture in order to produce your desired response.

Example:

- **Input:** Webcam
- **Detection:** Optical flow
- **Measurement:** Horizontal direction and velocity of movement
- **Mapping:** Map horizontal direction to stereo position; map velocity to volume
- **Target:** Audio playback in Max/MSP

Input: What sensor or device will capture the gesture?

Detection: How will the system recognize the action?

Measurement: What information do you need to extract?

Mapping: How will that information control the output?

Target: What will the measurement control?

3. TRACE THE TECHNICAL PIPELINE

Based on the demonstration and reference guide, identify which techniques you might need. You may not need every stage. Some inputs can be used directly without additional processing.

Stage	**Technique / Max object**	**Why do you need it?**
Input		
Pre-processing		
Detection / Tracking		
Analysis / Measurement		
Data Smoothing		
Parameter Mapping		
Target / Output		

4. WHAT DO YOU NEED TO FIGURE OUT?

What part of this interaction are you unsure how to implement?

What would you need to test to find out if your idea will work?

What is one simplified version of this interaction you could build as a first prototype?
